

Sub 1:

$$\sum_{n=1}^{+\infty} \frac{\sqrt[4]{n}}{\sqrt{n^4+1} + 2\sqrt[3]{n^3+1}}$$

$$x_n = \frac{n^{\frac{1}{4}}}{n^2 \sqrt{1+\frac{1}{n^4}} + 2n \sqrt[3]{1+\frac{1}{n^3}}} = \frac{1}{n^{\frac{7}{2}} \sqrt{1+\frac{1}{n^4}} + 2 \cdot n^{\frac{1}{2}} \sqrt[3]{1+\frac{1}{n^3}}}$$

Ver $\sum_{n=1}^{+\infty} g_n$, cu $(g_n)_{n \in \mathbb{N}^*}$ str descrescător, $g_n > 0$

$$g_n = \frac{1}{n^{\frac{7}{2}}}$$

$$\sum_{n=1}^{+\infty} g_n \rightarrow \text{serie armonică } \sum_{n=1}^{+\infty} \frac{1}{n^{\frac{7}{2}}}, \alpha = \frac{7}{2} > 1$$

$$\Rightarrow \sum_{n=1}^{+\infty} g_n \rightarrow \text{convergentă}$$

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{x_n}{g_n} &= \lim_{n \rightarrow \infty} \frac{1}{n^{\frac{7}{2}} \sqrt{1+\frac{1}{n^4}} + 2 \cdot n^{\frac{1}{2}} \sqrt[3]{1+\frac{1}{n^3}}} \cdot \frac{1}{\frac{1}{n^{\frac{7}{2}}}} \\ &= \lim_{n \rightarrow \infty} \frac{1}{n^{\frac{7}{2}} \sqrt{1+\frac{1}{n^4}} + 2 \cdot n^{\frac{1}{2}} \sqrt[3]{1+\frac{1}{n^3}}} \\ &= \lim_{n \rightarrow \infty} \frac{1}{\sqrt{1+\frac{1}{n^4}} + 2 \cdot n^{-1} \sqrt[3]{1+\frac{1}{n^3}}} = \frac{1}{\sqrt{1+0} + 2 \cdot 0 \sqrt[3]{1+0}} = \frac{1}{1} = 1 \end{aligned}$$

$$\Rightarrow \sum_{n=1}^{+\infty} x_n \sim \sum_{n=1}^{+\infty} g_n \quad (\text{seriile sunt de aceeași natură})$$

$$\sum_{n=1}^{+\infty} g_n \rightarrow \text{convergentă}$$

$$\Rightarrow \sum_{n=1}^{+\infty} x_n \rightarrow \text{convergentă}$$

Ex 6.2 $f: \mathbb{R}^2 \rightarrow \mathbb{R}$

$$f(x, y) = x^4 + y^3 - 4x^3 - 3y^2 + 3y$$

$$f \text{ cont. pe } \mathbb{R}^2 \text{ (op. a fctii. elem.)} \Rightarrow \Delta f = \emptyset$$

$$\frac{\partial f}{\partial x}(x, y) = 4x^3 - 12x^2, \quad \forall (x, y) \in \mathbb{R}^2$$

$$\frac{\partial f}{\partial y}(x, y) = 3y^2 - 6y + 3, \quad \forall (x, y) \in \mathbb{R}^2$$

$$\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \rightarrow \text{fctii. cont.}$$

$$\mathbb{R}^2 \rightarrow \text{multime deschis} \Rightarrow f \text{ diferentiaabil pe } \mathbb{R}^2 \\ \Rightarrow \Delta_1 = \emptyset$$

$$\Rightarrow f \rightarrow \text{functie de clasă } C_1$$

$$(\mathbb{R}^2) \quad \begin{cases} \frac{\partial f}{\partial x}(x, y) = 0 \\ \frac{\partial f}{\partial y}(x, y) = 0 \end{cases} \Rightarrow \begin{cases} 4x^3 - 12x^2 = 0 \Leftrightarrow 4x^2(x - 3) = 0 \\ 3y^2 - 6y + 3 = 0 \Leftrightarrow 3(y - 1)^2 = 0 \Rightarrow y = 1 \end{cases} \begin{matrix} x=0 \\ x=3 \end{matrix}$$

$$(0, 1) \in \mathbb{R}^2$$

$$\cancel{(0, 3) \in \mathbb{R}^2}$$

$$(3, 1) \in \mathbb{R}^2$$

$$\Rightarrow C = \{(0, 1), (3, 1)\}$$

$$\frac{\partial^2 f}{\partial x^2}(x, y) = 12x^2 - 24x, \quad \forall (x, y) \in \mathbb{R}^2$$

$$\frac{\partial^2 f}{\partial x \partial y}(x, y) = 0, \quad \forall (x, y) \in \mathbb{R}^2$$

$$\frac{\partial^2 f}{\partial y \partial x}(x, y) = 0, \quad \forall (x, y) \in \mathbb{R}^2$$

$$\frac{\partial^2 f}{\partial y^2}(x, y) = 6y - 6, \quad \forall (x, y) \in \mathbb{R}^2$$

$$\frac{\partial^2 f}{\partial x^2}, \frac{\partial^2 f}{\partial x \partial y}, \frac{\partial^2 f}{\partial y \partial x}, \frac{\partial^2 f}{\partial y^2} \rightarrow \text{fctii. cont.}$$

$$f \text{ admite toate derivatele de ordin 2} \Rightarrow \Delta_2 = \emptyset$$

$$\Rightarrow f \rightarrow \text{functie de clasă } C^2$$

$$Hf(x,y) = \begin{pmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{pmatrix} = \begin{pmatrix} 12x^2 - 24x & 0 \\ 0 & 6y - 6 \end{pmatrix}$$

$$Hf(0,1) = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \Rightarrow \begin{matrix} \Delta_1 = 0 \\ \Delta_2 = 0 \end{matrix} \Rightarrow \text{nu se poate decide}$$

$$Hf(3,1) = \begin{pmatrix} 36 & 0 \\ 0 & 0 \end{pmatrix} \Rightarrow \begin{matrix} \Delta_1 = 36 > 0 \\ \Delta_2 = 0 \end{matrix} \Rightarrow \text{nu se poate decide}$$

Trebuie folosită definiția

$$f(x,y) - f(0,1) = x^4 + y^3 - 4x^3 - 3y^2 + 3y - 1$$

$$= \cancel{x^3(x-4)}$$

$$= x^3(x-4) + y(y^2 - 3y + 1) - 1$$

$$f(0,0) - f(0,1) = -1 < 0$$

$$f(5,0) - f(0,1) = 124 > 0 \Rightarrow (0,1) \text{ nu este punct de extrem}$$

$$f(x,y) - f(3,1) = x^4 + y^3 - 4x^3 - 3y^2 + 3y + 26$$

$$f(0,0) - f(3,1) = 26 > 0$$

$$f(1,2) - f(3,1) = -3 - 26 + 26 = -3 < 0 \Rightarrow (3,1) \text{ nu este punct de extrem}$$

Sub 3

$$x_n = n(-1)^{n+1} \ln\left(1 + \frac{1}{2n}\right) + \cos \frac{n\pi}{2}$$

$$(-1)^{n+1} = \begin{cases} -1, & n=2k \\ 1, & n=2k+1 \end{cases}, k \in \mathbb{N}$$

$$\cos \frac{n\pi}{2} = \begin{cases} 0, & 2k+1 \\ 1, & 4k \\ -1, & 4k+2 \end{cases}, k \in \mathbb{N}$$

avem subșirurile $(x_{2k+1})_{k \in \mathbb{N}}$, $(x_{4k})_{k \in \mathbb{N}}$, $(x_{4k+2})_{k \in \mathbb{N}}$

$$\begin{aligned} \lim_{k \rightarrow \infty} x_{2k+1} &= \lim_{k \rightarrow \infty} (2k+1) \cdot \ln\left(1 + \frac{1}{4k+2}\right) + 0 \\ &= \lim_{k \rightarrow \infty} \ln\left(1 + \frac{1}{4k+2}\right)^{2k+1} \\ &= \ln\left(\lim_{k \rightarrow \infty} \left(1 + \frac{1}{4k+2}\right)^{\frac{2k+1}{4k+2}}\right) \\ &= \ln e^{\lim_{k \rightarrow \infty} \frac{2k+1}{4k+2}} = \ln e^{\frac{1}{2}} = \frac{1}{2} \end{aligned}$$

$$\begin{aligned} \lim_{k \rightarrow \infty} x_{4k} &= \lim_{k \rightarrow \infty} -4k \ln\left(1 + \frac{1}{8k}\right) + 1 \\ &= -\ln\left(\lim_{k \rightarrow \infty} \left(1 + \frac{1}{8k}\right)^{\frac{4k}{8k}}\right) + 1 \\ &= -\ln e^{\frac{1}{2}} + 1 = -\frac{1}{2} + 1 = \frac{1}{2} \end{aligned}$$

$$\begin{aligned} \lim_{k \rightarrow \infty} x_{4k+2} &= \lim_{k \rightarrow \infty} -(4k+2) \ln\left(1 + \frac{1}{8k+4}\right) - 1 \\ &= -\ln\left(\lim_{k \rightarrow \infty} \left(1 + \frac{1}{8k+4}\right)^{\frac{4k+2}{8k+4}}\right) - 1 \\ &= -\ln e^{\frac{1}{2}} - 1 = -\frac{1}{2} - 1 = -\frac{3}{2} \end{aligned}$$

$$\Rightarrow L = \left\{-\frac{3}{2}, \frac{1}{2}\right\}$$

$$\overline{\lim} x_n = \sup L \Rightarrow \overline{\lim} x_n = \frac{1}{2}$$

$$\underline{\lim} x_n = \inf L \Rightarrow \underline{\lim} x_n = -\frac{3}{2}$$

$$\iint_D x \, dx \, dy, \quad D = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 4; x \leq 0; y \geq 0\}$$

$$\begin{cases} x^2 + y^2 \leq 4 \\ x \leq 0 \\ y \geq 0 \end{cases} \Rightarrow y^2 \leq 4 - x^2 \Rightarrow -\sqrt{4-x^2} \leq y \leq \sqrt{4-x^2} \quad \uparrow \\ y \geq 0 \quad \uparrow \\ \Rightarrow 0 \leq y \leq \sqrt{4-x^2}$$

$$\Rightarrow 0 \leq \sqrt{4-x^2} \Rightarrow 0 \leq 4-x^2 \Rightarrow x^2 \leq 4 \Rightarrow -2 \leq x \leq 2 \quad \uparrow \\ x \leq 0 \quad \uparrow \\ \Rightarrow x \in [-2, 0]$$

$$0 \leq y \leq \sqrt{4-x^2}$$

$$g(x) \quad h(x)$$

$$g, h: [-2, 0] \rightarrow \mathbb{R}, \quad g(x) = 0, \quad h(x) = \sqrt{4-x^2}$$

$g, h \rightarrow$ func. continue

$$\Rightarrow D = \{(x, y) \in \mathbb{R}^2 \mid x \in [-2, 0], g(x) \leq y \leq h(x), g(x) = 0, h(x) = \sqrt{4-x^2}\}$$

$\Rightarrow D \rightarrow$ mul. (fi) me simple în raport cu xy $\Rightarrow$ ordinea de integrare este dy, dx

$$\begin{aligned} \Rightarrow \iint_D x \, dx \, dy &= \int_{-2}^0 \left(\int_{g(x)}^{h(x)} x \, dy \right) dx \\ &= \int_{-2}^0 \left(x y \Big|_0^{\sqrt{4-x^2}} \right) dx = \int_{-2}^0 (x \cdot \sqrt{4-x^2} - x \cdot 0) dx \\ &= \int_{-2}^0 x \sqrt{4-x^2} \, dx = -\frac{1}{2} \int_{-2}^0 -2x \sqrt{4-x^2} \, dx \\ &\quad \begin{matrix} t = 4-x^2 \\ dt = -2x \, dx \end{matrix} \\ &= -\frac{1}{2} \int_{\sqrt{4^3}}^{\frac{1}{2}} t^{\frac{1}{2}} \, dt = -\frac{1}{2} \cdot \frac{2}{3} t^{\frac{3}{2}} \Big|_0^{\sqrt{4^3}} \\ &= -\frac{\sqrt{4^3}}{3} + \frac{0^{\frac{3}{2}}}{3} = -\frac{8}{3} \end{aligned}$$